PANDAS

```
!pip list
```

```
cupy-cuda12x                      13.3.0
curl_cffi                         0.11.4
cuvs-cu12                         25.2.1
cvxopt                            1.3.2
cvxpy                             1.6.6
cycler                            0.12.1
cyipopt                           1.5.0
cymem                             2.0.11
Cython                            3.0.12
dask                              2024.12.1
dask-cuda                         25.2.0
dask-cudf-cu12                    25.2.2
dask-expr                         1.1.21
dataproc-spark-connect            0.7.5
datascience                       0.17.6
datasets                          3.6.0
db-dtypes                         1.4.3
dbus-python                       1.2.18
debugpy                           1.8.0
decorator                         4.4.2
defusedxml                        0.7.1
diffusers                         0.34.0
dill                              0.3.7
distributed                       2024.12.1
distributed-ucxx-cu12             0.42.0
distro                            1.9.0
dlib                              19.24.6
dm-tree                           0.1.9
docstring_parser                  0.16
docutils                          0.21.2
dopamine_rl                       4.1.2
duckdb                            1.2.2
earthengine-api                   1.5.22
easydict                          1.13
editdistance                      0.8.1
eerepr                            0.1.2
einops                            0.8.1
en_core_web_sm                    3.8.0
entrypoints                       0.4
et_xmlfile                        2.0.0
etils                             1.12.2
etuples                           0.3.9
Farama-Notifications              0.0.4
fastai                            2.7.19
fastapi                           0.115.14
fastcore                          1.7.29
fastdownload                      0.0.7
fastjsonschema                    2.21.1
fastprogress                      1.0.3
fastrlock                         0.8.3
ffmpy                             0.6.0
filelock                          3.18.0
firebase-admin                    6.9.0
Flask                             3.1.1
flatbuffers                       25.2.10
flax                              0.10.6
folium                            0.19.7
fonttools                         4.58.4
```

```
!pip install pandas
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.11/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.23.2 in /usr/local/lib/python3.11/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages (from pandas) (2025.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

```
import pandas as pd
```

```
df=pd.read_csv('/content/sample_data/california_housing_test.csv')

'''
csv nedir python da aldığım bir veri seti virgüllerle ayrılmışsa bu csv dosyasıdır.
'''
```

```
'\ncsv nedir python da aldığım bir veri seti virgüllerle ayrılmışsa bu csv dosyasıdır.\n'
```

```
type(df)
```

```
pandas.core.frame.DataFrame
def __init__(data=None, index: Axes | None=None, columns: Axes | None=None, dtype: Dtype |
None=None, copy: bool | None=None) -> None
```

```
Two-dimensional, size-mutable, potentially heterogeneous tabular data.

Data structure also contains labeled axes (rows and columns).
Arithmetic operations align on both row and column labels. Can be
thought of as a dict-like container for Series objects. The primary
pandas data structure.
```

```
df
```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_valu
0	-122.05	37.37	27.0	3885.0	661.0	1537.0	606.0	6.6085	344700.
1	-118.30	34.26	43.0	1510.0	310.0	809.0	277.0	3.5990	176500.
2	-117.81	33.78	27.0	3589.0	507.0	1484.0	495.0	5.7934	270500.
3	-118.36	33.82	28.0	67.0	15.0	49.0	11.0	6.1359	330000.
4	-119.67	36.33	19.0	1241.0	244.0	850.0	237.0	2.9375	81700.
...	...	...	...	...	...	...	...	...	.
2995	-119.86	34.42	23.0	1450.0	642.0	1258.0	607.0	1.1790	225000.
2996	-118.14	34.06	27.0	5257.0	1082.0	3496.0	1036.0	3.3906	237200.
2997	-119.70	36.30	10.0	956.0	201.0	693.0	220.0	2.2895	62000.
2998	-117.12	34.10	40.0	96.0	14.0	46.0	14.0	3.2708	162500.
2999	-119.63	34.42	42.0	1765.0	263.0	753.0	260.0	8.5608	500001.

3000 rows × 9 columns

```
df["total_rooms"]
```

	total_rooms
0	3885.0
1	1510.0
2	3589.0
3	67.0
4	1241.0
...	...
2995	1450.0
2996	5257.0
2997	956.0
2998	96.0
2999	1765.0

3000 rows × 1 columns

dtype: float64

```
df.total_rooms
```

	total_rooms
0	3885.0
1	1510.0
2	3589.0
3	67.0
4	1241.0
...	...
2995	1450.0
2996	5257.0
2997	956.0
2998	96.0
2999	1765.0

3000 rows × 1 columns

dtype: float64

```
df.total_bedrooms[0]
```

```
np.float64(661.0)
```

```
!pip install datasets #hugging face'in datasets kütüphanesidr
```

```
Requirement already satisfied: datasets in /usr/local/lib/python3.11/dist-packages (3.6.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from datasets) (3.18.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.11/dist-packages (from datasets) (2.0.2)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.11/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.11/dist-packages (from datasets) (0.3.7)
Requirement already satisfied: pandas in /usr/local/lib/python3.11/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.11/dist-packages (from datasets) (2.32.3)
Requirement already satisfied: tqdm>=4.66.3 in /usr/local/lib/python3.11/dist-packages (from datasets) (4.67.1)
Requirement already satisfied: xxhash in /usr/local/lib/python3.11/dist-packages (from datasets) (3.5.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.11/dist-packages (from datasets) (0.70.15)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.11/dist-packages (from fsspec[http]<=2025.3.0,>
Requirement already satisfied: huggingface-hub>=0.24.0 in /usr/local/lib/python3.11/dist-packages (from datasets) (0.33.2)
Requirement already satisfied: packaging in /usr/local/lib/python3.11/dist-packages (from datasets) (24.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.11/dist-packages (from datasets) (6.0.2)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.11/dist-packages (from fsspec[http]<=2025.3.0,>=
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub>=0.24.0->
Requirement already satisfied: hf-xet<2.0.0,>=1.1.2 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub>=0.24.0->datase
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->datasets
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->datasets) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->datasets) (2.4
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->datasets) (2025
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11/dist-packages (from pandas->datasets) (2.9.0.post
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: aiohappyeyeballs>=2.3.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1-
Requirement already satisfied: aiosignal>=1.1.2 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[htt
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fssp
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (from python-dateutil>=2.8.2->pandas->datasets)
```

```
from datasets import load_dataset
```

```
load_dataset('lukebarousse/data_jobs')
```

```
/usr/local/lib/python3.11/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as s
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
README.md:        3.25k/?  [00:00<00:00,  90.2kB/s]

data_jobs.csv:  100%                                                    231M/231M  [00:01<00:00,  140MB/s]

Generating train split:  100%                                                    785741/785741  [00:08<00:00,  68380.49  examples/s]
DatasetDict({
    train: Dataset({
        features: ['job_title_short', 'job_title', 'job_location', 'job_via', 'job_schedule_type', 'job_work_from_home',
'search_location', 'job_posted_date', 'job_no_degree_mention', 'job_health_insurance', 'job_country', 'salary_rate',
'salary_year_avg', 'salary_hour_avg', 'company_name', 'job_skills', 'job_type_skills'],
        num_rows: 785741
    })
})
```

```
pip install --upgrade huggingface_hub
```

```
Requirement already satisfied: huggingface_hub in /usr/local/lib/python3.11/dist-packages (0.33.2)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from huggingface_hub) (3.18.0)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.11/dist-packages (from huggingface_hub) (2025.3.0)
Requirement already satisfied: packaging>=20.9 in /usr/local/lib/python3.11/dist-packages (from huggingface_hub) (24.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.11/dist-packages (from huggingface_hub) (6.0.2)
Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages (from huggingface_hub) (2.32.3)
Requirement already satisfied: tqdm>=4.42.1 in /usr/local/lib/python3.11/dist-packages (from huggingface_hub) (4.67.1)
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.11/dist-packages (from huggingface_hub) (4.14.0
Requirement already satisfied: hf-xet<2.0.0,>=1.1.2 in /usr/local/lib/python3.11/dist-packages (from huggingface_hub) (1.1.5)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests->huggingface_hub)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests->huggingface_hub) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests->huggingface_hub) (2.4.6
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests->huggingface_hub) (2025
```

```
!pip install --upgrade datasets
#upgrade  install ettiğim yüklenmişse günceller yüklenmemişse paketi ilk defa yükler
```

```
Requirement already satisfied: datasets in /usr/local/lib/python3.11/dist-packages (3.6.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (from datasets) (3.18.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.11/dist-packages (from datasets) (2.0.2)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.11/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.11/dist-packages (from datasets) (0.3.7)
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Requirement already satisfied: tqdm>=4.66.3 in /usr/local/lib/python3.11/dist-packages (from datasets) (4.67.1)
Requirement already satisfied: xxhash in /usr/local/lib/python3.11/dist-packages (from datasets) (3.5.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.11/dist-packages (from datasets) (0.70.15)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.11/dist-packages (from fsspec[http]<=2025.3.0,>
Requirement already satisfied: huggingface-hub>=0.24.0 in /usr/local/lib/python3.11/dist-packages (from datasets) (0.33.2)
Requirement already satisfied: packaging in /usr/local/lib/python3.11/dist-packages (from datasets) (24.2)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.11/dist-packages (from datasets) (6.0.2)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.11/dist-packages (from fsspec[http]<=2025.3.0,>=
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub>=0.24.0->
Requirement already satisfied: hf-xet<2.0.0,>=1.1.2 in /usr/local/lib/python3.11/dist-packages (from huggingface-hub>=0.24.0->datase
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->datasets
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->datasets) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->datasets) (2.4
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests>=2.32.2->datasets) (2025
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11/dist-packages (from pandas->datasets) (2.9.0.post
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: aiohappyeyeballs>=2.3.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1-
Requirement already satisfied: aiosignal>=1.1.2 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[htt
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fssp
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (from python-dateutil>=2.8.2->pandas->datasets)
```

```
dataset=load_dataset('lukebarousse/data_jobs')
```

```
dataset
```

```
DatasetDict({
    train: Dataset({
        features: ['job_title_short', 'job_title', 'job_location', 'job_via', 'job_schedule_type', 'job_work_from_home',
'search_location', 'job_posted_date', 'job_no_degree_mention', 'job_health_insurance', 'job_country', 'salary_rate',
'salary_year_avg', 'salary_hour_avg', 'company_name', 'job_skills', 'job_type_skills'],
        num_rows: 785741
    })
})
```

```
dataset["train"]
```

```
Dataset({
    features: ['job_title_short', 'job_title', 'job_location', 'job_via', 'job_schedule_type', 'job_work_from_home',
'search_location', 'job_posted_date', 'job_no_degree_mention', 'job_health_insurance', 'job_country', 'salary_rate',
'salary_year_avg', 'salary_hour_avg', 'company_name', 'job_skills', 'job_type_skills'],
    num_rows: 785741
})
```

```
type(dataset["train"])
```

```
datasets.arrow_dataset.Dataset
def __init__(arrow_table: Table, info: Optional[DatasetInfo]=None, split: Optional[NamedSplit]=None,
indices_table: Optional[Table]=None, fingerprint: Optional[str]=None)
```

```
A Dataset backed by an Arrow table.
```

```
df=dataset["train"].to_pandas() #burada neden read_csv metodunu  kullanmadık çünkü burada verim zaten normal düzgün bi veri bi dosya
#şeklinde değil o yüzden to_pandas metodunu kullanıyorum.
```

```
df
```

	job_title_short	job_title	job_location	job_via	job_schedule_type	job_work_from_home	search_location
0	Senior Data Engineer	Senior Clinical Data Engineer / Principal Clin...	Watertown, CT	via Work Nearby	Full-time	False	Texas, United States
1	Data Analyst	Data Analyst	Guadalajara, Jalisco, Mexico	via BeBee México	Full-time	False	Mexico
2	Data Engineer	Data Engineer/Scientist/Analyst, Mid or Senior...	Berlin, Germany	via LinkedIn	Full-time	False	Germany
3	Data Engineer	LEAD ENGINEER - PRINCIPAL ANALYST - PRINCIPAL ...	San Antonio, TX	via Diversity.com	Full-time	False	Texas, United States
4	Data Engineer	Data Engineer- Sr Jobs	Washington, DC	via Clearance Jobs	Full-time	False	Sudan
...	...	...	...	...	...	...	...
785736	Software Engineer	DevOps Engineer	Singapura	melalui Trabajo.org	Pekerjaan tetap	False	Singapore
785737	Data Analyst	CRM Data Analyst	Bad Rodach, Jerman	melalui BeBee Deutschland	Pekerjaan tetap	False	Germany
785738	Business Analyst	Commercial Analyst - Start Now	Malaysia	melalui Ricebowl	Pekerjaan tetap	False	Malaysia
785739	Data Engineer	Principal Associate, Data Engineer (Remote-Eli...	Newark, New Jersey, Amerika Serikat	melalui Recruit.net	Pekerjaan tetap	False	Sudan
785740	Software Engineer	AWS System Analyst	India	melalui Trigyn	Pekerjaan tetap	False	India

785741 rows × 17 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 785741 entries, 0 to 785740
Data columns (total 17 columns):
 #   Column                Non-Null Count   Dtype
---  ------                --------------   -----
 0   job_title_short       785741 non-null  object
 1   job_title             785740 non-null  object
 2   job_location          784696 non-null  object
```

```
 3   job_via                785733 non-null  object
 4   job_schedule_type      773074 non-null  object
 5   job_work_from_home     785741 non-null  bool
 6   search_location        785741 non-null  object
 7   job_posted_date        785741 non-null  object
 8   job_no_degree_mention  785741 non-null  bool
 9   job_health_insurance   785741 non-null  bool
 10  job_country            785692 non-null  object
 11  salary_rate            33067 non-null   object
 12  salary_year_avg        22003 non-null   float64
 13  salary_hour_avg        10662 non-null   float64
 14  company_name           785723 non-null  object
 15  job_skills             668704 non-null  object
 16  job_type_skills        668704 non-null  object
dtypes: bool(3), float64(2), object(12)
memory usage: 86.2+ MB
```

```
df
```

	job_title_short	job_title	job_location	job_via	job_schedule_type	job_work_from_home	search_location
0	Senior Data Engineer	Senior Clinical Data Engineer / Principal Clin...	Watertown, CT	via Work Nearby	Full-time	False	Texas, United States
1	Data Analyst	Data Analyst	Guadalajara, Jalisco, Mexico	via BeBee México	Full-time	False	Mexico
2	Data Engineer	Data Engineer/Scientist/Analyst, Mid or Senior...	Berlin, Germany	via LinkedIn	Full-time	False	Germany
3	Data Engineer	LEAD ENGINEER - PRINCIPAL ANALYST - PRINCIPAL ...	San Antonio, TX	via Diversity.com	Full-time	False	Texas, United States
4	Data Engineer	Data Engineer- Sr Jobs	Washington, DC	via Clearance Jobs	Full-time	False	Sudan
...	...	...	...	...	...	...	...
785736	Software Engineer	DevOps Engineer	Singapura	melalui Trabajo.org	Pekerjaan tetap	False	Singapore
785737	Data Analyst	CRM Data Analyst	Bad Rodach, Jerman	melalui BeBee Deutschland	Pekerjaan tetap	False	Germany
785738	Business Analyst	Commercial Analyst - Start Now	Malaysia	melalui Ricebowl	Pekerjaan tetap	False	Malaysia
785739	Data Engineer	Principal Associate, Data Engineer (Remote-Eli...	Newark, New Jersey, Amerika Serikat	melalui Recruit.net	Pekerjaan tetap	False	Sudan
785740	Software Engineer	AWS System Analyst	India	melalui Trigyn	Pekerjaan tetap	False	India

785741 rows × 17 columns

```
df.head() #data framin ilk 5 satırını görmenizi sağlar,böylelikle eksik verileri hemencecik görmemizi sağlar.
#daha fazla satır görmek istiyorsak,parantez içinde sayıyı yazabiliriz.
```

	job_title_short	job_title	job_location	job_via	job_schedule_type	job_work_from_home	search_location	job_p
0	Senior Data Engineer	Senior Clinical Data Engineer / Principal Clin...	Watertown, CT	via Work Nearby	Full-time	False	Texas, United States	
1	Data Analyst	Data Analyst	Guadalajara, Jalisco, Mexico	via BeBee México	Full-time	False	Mexico	
2	Data Engineer	Data Engineer/Scientist/Analyst, Mid or Senior...	Berlin, Germany	via LinkedIn	Full-time	False	Germany	
3	Data Engineer	LEAD ENGINEER - PRINCIPAL ANALYST - PRINCIPAL ...	San Antonio, TX	via Diversity.com	Full-time	False	Texas, United States	
4	Data Engineer	Data Engineer- Sr Jobs	Washington, DC	via Clearance Jobs	Full-time	False	Sudan	

```
df.head(7)
```

	job_title_short	job_title	job_location	job_via	job_schedule_type	job_work_from_home	search_location	job_p
0	Senior Data Engineer	Senior Clinical Data Engineer / Principal Clin...	Watertown, CT	via Work Nearby	Full-time	False	Texas, United States	
1	Data Analyst	Data Analyst	Guadalajara, Jalisco, Mexico	via BeBee México	Full-time	False	Mexico	
2	Data Engineer	Data Engineer/Scientist/Analyst, Mid or Senior...	Berlin, Germany	via LinkedIn	Full-time	False	Germany	
3	Data Engineer	LEAD ENGINEER - PRINCIPAL ANALYST - PRINCIPAL ...	San Antonio, TX	via Diversity.com	Full-time	False	Texas, United States	
4	Data Engineer	Data Engineer- Sr Jobs	Washington, DC	via Clearance Jobs	Full-time	False	Sudan	
5	Data Engineer	GCP Data Engineer	Anywhere	via ZipRecruiter	Contractor and Temp work	True	Georgia	
6	Senior Data Engineer	Senior Data Engineer - GCP Cloud	Dearborn, MI	via LinkedIn	Full-time	False	Florida, United States	

```
df.tail()  #dataframin son bikaç satırını görmek için kullanılır.
```

	job_title_short	job_title	job_location	job_via	job_schedule_type	job_work_from_home	search_location	job_posted_da
785736	Software Engineer	DevOps Engineer	Singapura	melalui Trabajo.org	Pekerjaan tetap	False	Singapore	2023-03-06:16:
785737	Data Analyst	CRM Data Analyst	Bad Rodach, Jerman	melalui BeBee Deutschland	Pekerjaan tetap	False	Germany	2023-03-06:18:
785738	Business Analyst	Commercial Analyst - Start Now	Malaysia	melalui Ricebowl	Pekerjaan tetap	False	Malaysia	2023-03-06:32:
785739	Data Engineer	Principal Associate, Data Engineer (Remote-Eli...	Newark, New Jersey, Amerika Serikat	melalui Recruit.net	Pekerjaan tetap	False	Sudan	2023-03-06:32:
785740	Software Engineer	AWS System Analyst	India	melalui Trigyn	Pekerjaan tetap	False	India	2023-03-06:16:

```
df.tail(10) #kaç tane son satırlardan görmek istiyorum.
```

	job_title_short	job_title	job_location	job_via	job_schedule_type	job_work_from_home	search_location	job_posted_da
785731	Data Engineer	Data Engineer	Overland Park, Kansas, Amerika Serikat	melalui LinkedIn	Pekerjaan tetap	False	Sudan	2023-03-06:32:
785732	Data Engineer	Data Engineer (f/m/d)	Heidelberg, Jerman	melalui Top County Careers	Pekerjaan tetap	False	Germany	2023-03-06:18:
785733	Software Engineer	DevOps Engineer	Singapura	melalui Trabajo.org	Pekerjaan tetap	False	Singapore	2023-03-06:16:
785734	Business Analyst	PreSales Engineer	Almaty, Kazakhstan	melalui Melga	Pekerjaan tetap	False	Kazakhstan	2023-03-06:31:
785735	Senior Data Engineer	Senior Data Engineer	Berlin, Jerman	melalui Top County Careers	Pekerjaan tetap	False	Germany	2023-03-06:19:
785736	Software Engineer	DevOps Engineer	Singapura	melalui Trabajo.org	Pekerjaan tetap	False	Singapore	2023-03-06:16:
785737	Data Analyst	CRM Data Analyst	Bad Rodach, Jerman	melalui BeBee Deutschland	Pekerjaan tetap	False	Germany	2023-03-06:18:
785738	Business Analyst	Commercial Analyst - Start Now	Malaysia	melalui Ricebowl	Pekerjaan tetap	False	Malaysia	2023-03-06:32:
785739	Data Engineer	Principal Associate, Data Engineer (Remote-Eli...	Newark, New Jersey, Amerika Serikat	melalui Recruit.net	Pekerjaan tetap	False	Sudan	2023-03-06:32:
785740	Software Engineer	AWS System Analyst	India	melalui Trigyn	Pekerjaan tetap	False	India	2023-03-06:16:

```
df['job_title_short'].head()
```

	job_title_short
0	Senior Data Engineer
1	Data Analyst
2	Data Engineer
3	Data Engineer
4	Data Engineer

dtype: object

```
#daha fazla sütun görmek istersem
df[['job_title_short', 'job_location']]
```

	job_title_short	job_location
0	Senior Data Engineer	Watertown, CT
1	Data Analyst	Guadalajara, Jalisco, Mexico
2	Data Engineer	Berlin, Germany
3	Data Engineer	San Antonio, TX
4	Data Engineer	Washington, DC
...	...	...
785736	Software Engineer	Singapura
785737	Data Analyst	Bad Rodach, Jerman
785738	Business Analyst	Malaysia
785739	Data Engineer	Newark, New Jersey, Amerika Serikat
785740	Software Engineer	India

785741 rows × 2 columns

```
df[['job_title_short', 'job_location']].head()
```

	job_title_short	job_location
0	Senior Data Engineer	Watertown, CT
1	Data Analyst	Guadalajara, Jalisco, Mexico
2	Data Engineer	Berlin, Germany
3	Data Engineer	San Antonio, TX
4	Data Engineer	Washington, DC

```
df.iloc[90]#dataframin 90.satırındaki  verilere erişmek için kullanılır.
```

	90
job_title_short	Data Scientist
job_title	Data Scientist
job_location	Australia
job_via	via BeBee Australia
job_schedule_type	Full-time
job_work_from_home	False
search_location	Australia
job_posted_date	2023-11-25 13:12:03
job_no_degree_mention	False
job_health_insurance	False
job_country	Australia
salary_rate	None
salary_year_avg	NaN
salary_hour_avg	NaN
company_name	Concurrent Technologies Corporation
job_skills	['python', 'sql', 'r', 'java', 'bash', 'react'...
job_type_skills	{'analyst_tools': ['tableau'], 'libraries': ['...

dtype: object

```
df.iloc[90:93]#90 ile  93 arasındaki verileri döndürmek için kullanılır.
```

	job_title_short	job_title	job_location	job_via	job_schedule_type	job_work_from_home	search_location	job_posted_date	job_
90	Data Scientist	Data Scientist	Australia	via BeBee Australia	Full-time	False	Australia	2023-11-25 13:12:03	
91	Data Scientist	Data Scientist	Temuco, Chile	via BeBee Chile	Full-time	False	Chile	2023-01-06 14:07:40	
92	Data Engineer	Remote - Data Engineer - Permanent - W2	Anywhere	via LinkedIn	Full-time	True	Illinois, United States	2023-02-21 13:29:59	

```
df.iloc[90:93,2]#90 ile 93 arasındaki verilerin 2 sütunun getirir.
```

	job_location
90	Australia
91	Temuco, Chile
92	Anywhere

dtype: object

```
df.iloc[90:93,0:2]
```

	job_title_short	job_title
90	Data Scientist	Data Scientist
91	Data Scientist	Data Scientist
92	Data Engineer	Remote - Data Engineer - Permanent - W2

```
df[['job_title_short','job_title']].iloc[90:93]
```

	job_title_short	job_title
90	Data Scientist	Data Scientist
91	Data Scientist	Data Scientist
92	Data Engineer	Remote - Data Engineer - Permanent - W2

```
df.job_location.unique()#bir sütundaki benzersiz verileri getirir.
```

```
array(['Watertown, CT', 'Guadalajara, Jalisco, Mexico', 'Berlin, Germany',
       ..., 'Heidelberg, Jerman', 'Bad Rodach, Jerman',
       'Newark, New Jersey, Amerika Serikat'], dtype=object)
```

```
df.job_location=='Germany'
```

	job_location
0	False
1	False
2	False
3	False
4	False
...	...
785736	False
785737	False
785738	False
785739	False
785740	False

785741 rows × 1 columns

dtype: bool

```
#yukarıda yazdığım kodu df parantezine alırsam pandas dataFrame dönüştürmüş olacağım
df[df.job_location=='Germany']
```

	job_title_short	job_title	job_location	job_via	job_schedule_type	job_work_from_home	search_location	job
1725	Data Engineer	Data Engineer SAP Reporting (m/w/d)	Germany	via Indeed	Full-time	False	Germany	
2177	Data Scientist	Data Scientist Kundendatenauswertung	Germany	via BeBee Deutschland	None	False	Germany	
2618	Data Engineer	Remote: Data Engineers	Germany	via BeBee	Full-time, Part-time, and Temp work	False	Germany	
2904	Data Engineer	Data Engineer (m/f/d)	Germany	via LinkedIn	Full-time	False	Germany	
3099	Senior Data Scientist	Senior Data Scientist Remote	Germany	via BeBee	Full-time	False	Germany	
...	...	...	...	...	...	...	...	
781829	Data Scientist	Data Scientist with Focus on Experience Analyses	Germany	via BeBee Deutschland	Full-time	False	Germany	
783250	Data Scientist	Data Scientist	Germany	via Jobs Trabajo.org	Full-time	False	Germany	
783636	Data Analyst	Data Analyst	Germany	via BeBee	Full-time	False	Germany	
783879	Data Scientist	Data Scientist	Germany	via BeBee	Full-time	False	Germany	
785168	Data Analyst	Data Analyst COO Business Management	Germany	via BeBee	Full-time	False	Germany	

2447 rows × 17 columns

```
df[(df.job_location=='Germany') & (df.salary_year_avg>100000)].head(5)
```

	job_title_short	job_title	job_location	job_via	job_schedule_type	job_work_from_home	search_location	job_posted_date
48733	Software Engineer	(Senior) Web Analytics Engineer (m/f/d)	Germany	via Ai-Jobs.net	Full-time	False	Germany	2023-12-14 06:15:50
51100	Data Analyst	(Senior) People Data Analyst (m/w/x)	Germany	via Ai-Jobs.net	Full-time	False	Germany	2023-05-15 07:18:25
162387	Data Scientist	Data Scientist (all genders)	Germany	via Ai-Jobs.net	Full-time	False	Germany	2023-11-10 15:35:24
195322	Data Engineer	Data Engineer* / Software Engineer* im Bereich...	Germany	via Ai-Jobs.net	Full-time	False	Germany	2023-12-09 15:12:20
210485	Machine Learning Engineer	Machine Learning Engineer-Search and Recommen...	Germany	via Ai-Jobs.net	Full-time	False	Germany	2023-01-11 22:23:59

```
df[(df.job_location=='Germany') & (df.salary_year_avg.notna())]#eğer sütunumda boş bi değer veya geçersiz bi değer varsa false döndürür
#değilse true döndürür
```